

Honeywell TC Thermostat Configuration Wizard

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ABOUT THIS GUIDE

This document provides instructions to configure the TC Thermostat Configuration Wizard connected to the Niagara workbench.

To take full advantage of the information in this guide, users must have training or experience working with the Niagara N4 software.

PREREQUISITE

Table 1 Prerequisite

Parameter	Description
Software requirement	Niagara workbench. For compatibility information, refer to the latest software releaser notes.
	Google Chrome web browser
Modules	• honeywellTCThermostatWizard-rt.jar • honeywellTCThermostatWizard-ux.jar • honeywellTCThermostatWizard-wb.jar • honeywellBacnetDeviceManager-rt.jar • honeywellBacnetDeviceManager-wb.jar • honeywellDeviceManager-rt.jar • honeywellDeviceManager-wb.jar
Unit Settings	The wizard supports Imperial and Metric measurement units. To change the units in the wizard, follow the below steps. 1. Navigate to the user service of the station. 2. Double-click on the user account and configure the Unit Conversion property. If the Unit Conversion is set to None, the Imperial units are applied.

INTRODUCTION

The Honeywell TC Thermostat Configuration Wizard is a utility that provides a systematic approach for configuring the various operational sequences and settings for the TC500A and TC300B thermostats. The wizard configures a TC500A thermostat for conventional, Air source, Water Source heat pump and Fan coil unit and a TC300B thermostat for Fan coil unit.

It can be accessed using a workbench or web browser. For more details, refer to the TC Thermostat Configuration Wizard User Guide - 31-00490.



NOTE:

Before proceeding with the configuration upload or download, update the thermostat firmware to version TC500A = 01.01.05.00 and TC300 = 01.02.09.08 or higher to ensure compatibility between the wizard and the thermostat.

Launching the Wizard

Follow the instructions below to launch the TC500A thermostat configuration wizard:

1. Navigate to the TC500A device, click on **Station > Config > Drivers > BacnetNetwork > TC500A**.
2. Double-click on TC500A to launch the TC500A Thermostat configuration wizard application.

Schedules

TC500A supports the schedule for occupancy and purge modes.

Table 2 Schedules

Schedule Name	Description	Type	BACnet Object ID
OccSchedule	Monday - Friday (6:00 AM TO 6:00 PM) 0: Occupied 1: Unoccupied 3: StandBy 255: Null	Enum Schedule	2
Purge Schedule	0: PurgeOff 1: PurgeOn The default is PurgeOff for all 5 days.	Enum Schedule	2

The schedules are added to the TC500A device by default. Configure the Occupancy and **PurgeControlSchedule** to change the modes and schedule the time. If a global schedule is present, link it to the **OccSchedule** and the **PurgeSchedule** using the **BacnetScheduleExport** component.

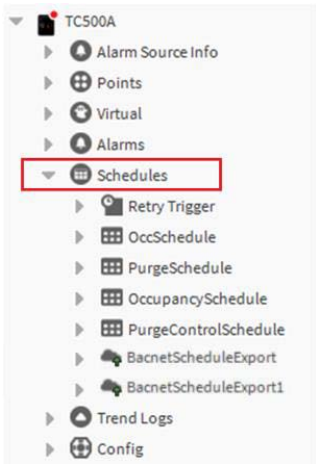


Fig. 1 Schedules

Follow the below steps to link the global schedule instead of the given occupancy schedule:

1. Navigate to the **BacnetScheduleExport** component and update the supervisor ord with the required schedule.

Property Sheet

BacnetScheduleExport (Bacnet Schedule Export)

Status	{fault}
State	Idle
Enabled	☒ true
Execution Time	5 minutes {Sun Mon Tue Wed Thu Fri Sat}
Last Attempt	18-Nov-2022 09:43:43 PM IST
Last Success	null
Last Failure	18-Nov-2022 09:43:43 PM IST
Fault Cause	<pre>java.lang.UnsupportedOperationException</pre>
Supervisor Ord	station: slot:/Drivers/BacnetNetwork/TC500/schedules/Occupan
Object Id	schedule 2
Data Type	Unsigned
Priority For Writing	15 [1-16]
Skip Writes	☒ ☐
Write Enum As	Unsigned
Out Of Service	☒ false

Refresh Save

Fig. 2 BacnetScheduleExport Property Sheet

2. Click Save

Field Description

TITLE BAR

NAVIGATION PANE

MAIN WINDOW

ACTION BUTTONS

Honeywell | TC500A Thermostat Configuration Wizard

General Equipment Control Loops Setpoints PID Config Fan Modes Sensors Terminal Assignment

General

Device name

Date

Time

Firmware version

SAVE BUILD POINTS

Fig. 3 TC500A Thermostat Configuration Wizard

TC Thermostat Configuration Wizard Application View

Title Bar

Displays the application name as **TC Thermostat Configuration Wizard**.

Navigation Pane

Enables the user to navigate through the various configurations of the equipment, sensors, PID control, etc.

See [Navigation Menu on page 8](#)

Main Window

It provides configuration settings based on the chosen parameters.

See [General - TC500A on page 9](#)

Action Buttons

- **SAVE:** Enables the user to save the changes made to a configuration.
- **BUILD POINTS:** Enables the user to build the points in the points folder of the TC500 device. See Table 2 below for the list of points built after clicking on the build points button.



NOTE:

Build Points are not applicable for TC300 Thermostat.

Table 3 Build Points

Point Name	Range	Type	BACnet Object ID
Common Points for Conventional and Heat Pump Applications			
no_EffSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Output	5
no_ActiveHeatStages	0 to 3	BACnet Numeric Output	7
no_ActiveCoolStages	0 to 3	BACnet Numeric Output	11
no_DaTemp	35 to 165 °F (1 to 73 °C)	BACnet Numeric Output	13
no_OaTemp	-40 to 150 °F (-39 to 65 °C)	BACnet Numeric Output	16
no_SpaceTemp	-40 to 140 °F (-39 to 60 °C)	BACnet Numeric Output	18
no_SpaceHumidity	0 to 100 %	BACnet Numeric Output	19
Cfg_Setpoints_OccCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	4
Cfg_Setpoints_UnOccCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	6
Cfg_Setpoints_OccHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	7
Cfg_Setpoints_UnOccHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	9
Cfg_Thermostat_ClAdjStPt	-30 to 30 °F (-34 to -1 °C)	BACnet Numeric Value	256
Cfg_Thermostat_HtAdjStPt	-30 to 30 °F (-34 to -1 °C)	BACnet Numeric Value	257
no_OccupancyState	0: Unoccupied 1: Occupied	BACnet Boolean Output	3
no_FanStart	0: Off 1: On	BACnet Boolean Output	19

Table 3 Build Points (Continued)

Point Name	Range	Type	BACnet Object ID
no_EffTempMode	1: Cool Mode 2: Reheat Mode 3: Heat Mode 4: Emergency Heat 5: Off	BACnet Enum Output	6
no_EffOccState	1: Occupied 2: Unoccupied 3: Bypass 4: Standby 5: Null	BACnet Enum Output	20
ni_OccManCom	1: Occupied 2: Unoccupied 3: Bypass 4: Standby 5: No Override	BACnet Enum Value	4
Cfg_Thermostat_SysSwitch	1: Auto 2: Cool 3: Heat 4: EmergHeat 5: Off	BACnet Enum Value	8
Cfg_Setpoints_StbyCoolSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	5
Cfg_Setpoints_StbyHeatSp	40 to 120 °F (4 to 48 °C)	BACnet Numeric Value	8
ni_SpaceCO2	0 to 2000 ppm	BACnet Numeric Value	81
ni_OutsideHum	0 to 100 %	BACnet Numeric Value	194
Additional Points for Heat Pump Applications			
no_ReversingVlv	0: Close 1: Open	BACnet Boolean Output	16
no_ActiveCompHeatStages	0 to 3	BACnet Numeric Output	10
no_ActiveAuxHeatStages	0 to 2	BACnet Numeric Output	8

CONFIGURATION

Navigation Menu

General

The General Menu allows the user to configure the miscellaneous settings for the TC500A thermostat. See [General - TC500A on page 9](#) and See [General - TC300 on page 10](#)

Equipment

This Menu allows the user to select the equipment type and subtypes. Any configuration change will generate a unique template number that determines the points to be built when the Build Points button is clicked. See [Equipment - TC500A on page 11](#) and See [Equipment - TC300 on page 12](#)

Control Loops

This Menu allows the user to select the Economizer status, Damper positions, CO2 control, Humidification and Dehumidification. Any change in configuration results in heating and cooling. Click the build points button to generate the points associated with the template number. See [Control Loops - TC500A on page 13](#).

**NOTE:**

Applicable only for TC500A Device.

Setpoints

This Menu allows the user to configure the setpoints for different modes like occupied, standby, and unoccupied. See [Setpoints - TC500A on page 14](#) and See [Setpoints - TC300 on page 15](#)

PID Configuration

This Menu allows the user to configure the PID parameters for heating and cooling modes. See [PID Configurations - TC500A on page 16](#) and See [PID Configurations - TC300 on page 17](#)

Fan Modes

This Menu allows the user to configure the different fan modes, speed types, and other fan settings. See [Fan Modes - TC500A on page 18](#) and See [Fan Type - TC300 on page 19](#)

Sensors

This Menu allows the user to configure the different types of sensors present in a system as well as the main control sensor. See [Sensors - TC500A on page 20](#) and See [Sensors - TC300 on page 21](#)

Terminal Assignment

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings. See [Terminal Assignment - TC500A on page 22](#) and See [on page 23](#)

GENERAL - TC500A

The general Menu displays the thermostat properties such as the Device name, Location, Date, Time, Firmware version, Wi-Fi signal strength, MSTP header CRC error count, and MSTP data CRC error count. It also has miscellaneous properties such as the Power up delay time, Override time, and Demand response setpoint adjustment.

Honeywell | TC500A Thermostat Configuration Wizard Configuration wizard ▼

General | Equipment | Control Loops | Setpoints | PID Config. | Fan Modes | Sensors | Terminal Assignment

General

Device name

Date

Time

Firmware version

Wi-Fi signal strength

MSTP header CRC error count 0.0

MSTP data CRC error count 0.0

Miscellaneous

Power up delay time s[0.0-300.0]

Override time min[0.0-1080.0]

Demand response setpoint adjustment °F[0.0-10.0]

System mode ▼

System mode changeover ☒ Auto ☐ Manual

Smoke mode ▼

System mode config ▼

SAVE **BUILD POINTS**

Fig. 4 General Menu View



NOTE:

Click **SAVE** for the configuration changes to reflect on the wizard. If the device is offline, not all of the parameters are displayed.

GENERAL - TC300

The general Menu displays the thermostat properties such as the Firmware Revision, MSTP header CRC error count, and MSTP data CRC error count, MSTP CRC Error Reset. It also has miscellaneous properties such as the Power up delay time, Override time, and Unoccupied Override Time, System Mode.

TC300 Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

General

Device Name

Date

Time

Firmware Revision

02.01.00.00

Model

TC300C-G

Thermostat Temperature Engineering Unit

Fahrenheit

☐ Sync HMI Display Unit with Engineering Unit

MSTP Header CRC Error Count

0

MSTP Data CRC Error Count

0

MSTP CRC Error Reset

Reset

Miscellaneous

Power Up Delay Time

10

s[0, 300]

Override Setting

☐ Permanent

☒ Temporary

Unoccupied Override Time

180

min[0, 1080]

Schedule

☒ Programmable commercial use

☐ Fixed Setpoints

System Mode

Auto

System Switch

☒ Auto

☒ Heat&Cool

☒ Fan Only

☒ Off

Fan Speed

Auto

Fan Mode Config

☒ Auto

☒ Adjustable/continuous

☒ Circulate

Valve Cycle

1 Minute Every 24 Hours

SAVE

Fig. 5 General Menu View

NOTE:

Click **SAVE** for the configuration changes to reflect on the wizard. If the device is offline, not all of the parameters are displayed.

EQUIPMENT - TC500A

This Menu allows the user to select the equipment type and subtype. Any change in configuration results in generating an unique application selection number. Click the build points button to generate the points associated with the template number.

Equipment Type

Select the required equipment type through this option. It is a primary setting in the configuration as selecting the various other parameters from the different locations depends on the equipment type. The additional settings for the equipment will be enabled or disabled according to the equipment type selected.

The following equipment types are available:

- Conventional
- Air source heat pump
- Water source heat pump.

Honeywell | TC500A Thermostat Configuration Wizard Configuration wizard ▾

General **Equipment** Control Loops Setpoints PID Config. Fan Modes Sensors Terminal Assignment

Equipment
Equipment type: Conventional ▾

Cooling

Cooling type: ☐ None ☒ Staged ☐ Modulating

Cooling stages: 2 stages ▾
1 stage
2 stages
3 stages
4 stages
Undefined5
Undefined6
Undefined7
Undefined8
Undefined9
Undefined10
Undefined11
Undefined12
Undefined13
Undefined14
Undefined15
Undefined16

Cooling minimum off time: [0-300.0]

Cooling minimum on time: [0-300.0]

Cooling cycles per hour: [0]

Heating

Heating type: ☒ Staged ☐ Modulating

Heating stages: [0]

Heating minimum off time: 60.0 s[0.0-300.0]

Heating minimum on time: 120.0 s[0.0-300.0]

Heating cycles per hour: 6.0 [2-20]

Heat fuel type: Standard efficiency gas ▾

SAVE **BUILD POINTS**

Fig. 6 Equipment Type

EQUIPMENT - TC300

This Menu allows the user to select the equipment type and subtype. Any change in configuration results in generating an unique application selection number. Click the build points button to generate the points associated with the template number.

Equipment Type

Select the required equipment type through this option. It is a primary setting in the configuration as selecting the various other parameters from the different locations depends on the equipment type. The additional settings for the equipment will be enabled or disabled according to the equipment type selected.

The following equipment types are available:

- 4 Pipe Dual Coil
- 4 Pipe Single Coil
- 2 Pipe Single Coil

Honeywell

| TC300 Thermostat Configuration Wizard

Configuration wizard ▾

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Equipment

Equipment Type

Fan Coil ▾

Fan Coil Type

4 Pipe Dual Coil ▾

Heating Valve Type

On/Off ▾

Heating On/Off Characteristic

☐ Normally Open

☒ Normally Closed

Cooling Valve Type

On/Off ▾

Cooling On/Off Characteristic

☐ Normally Open

☒ Normally Closed

Sensors

Drain Pan

☐

Aux Heat

Auxiliary Heat Enable

☐

Dehumidification

Dehumidification Enable

☐

SAVE

Fig. 7 Equipment Type

CONTROL LOOPS - TC500A

This Menu allows the user to select the Economizer status, Damper positions, CO2 control, Humidification and Dehumidification. Any change in configuration results in heating and cooling. Click the build points button to generate the points associated with the template number.

TC500A Thermostat Configuration Wizard

Configuration wizard ▼

General
Equipment
Control Loops
Setpoints
PID Config.
Fan Modes
Sensors
Terminal Assignment

Economizer ⓘ

Economizer status

None ▼

Damper positions ⓘ

Vent min. position for high fan speed

6.0

%[0.0-100.0]

CO2 control

CO2 output

☐ 0-10V
☒ 2-10V

Purge ⓘ

Purge enable

☐ Enable
☒ Disable

Humidification

Space relative low humidity setpoint

35.0

%[0.0-100.0]

Humidification on delay

20.0

min[0.0-60.0]

Dehumidification

Space relative high humidity setpoint

65.0

%[0.0-100.0]

Dehumidification on delay

20.0

min[0.0-60.0]

SAVE

BUILD POINTS

Fig. 8 Control Loops

Honeywell

TC500A Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Control Loops

Setpoints

PID Config.

Fan Modes

Sensors

Terminal Assignment

Setpoints

Unoccupied heat

55.0

°F[40.0-90.0]

Standby heat

65.0

°F[40.0-90.0]

Occupied heat

68.0

°F[40.0-90.0]

Occupied cool

76.0

°F[50.0-99.0]

Standby cool

80.0

°F[50.0-99.0]

Unoccupied cool

85.0

°F[50.0-99.0]

Setpoint stops

Minimum cooling setpoint

50.0

°F[50.0-99.0]

Maximum heating setpoint

90.0

°F[40.0-90.0]

Setpoint limit

Thermostat deadband

3.0

°F[2.0-8.0]

Temporary setpoint limit

30.0

°F[0.0-30.0]

Cooling setpoint recovery

Maximum cooling setpoint ramp

6.0

°F/hr[0.0-20.0]

OAT at maximum cooling setpoint ramp

70.0

°F[-40.0-120.0]

Minimum heating setpoint ramp

2.0

°F/hr[0.0-36.0]

OAT at minimum heating setpoint ramp

0.0

°F[-40.0-120.0]

DeltaT configurations

Staged heating

☐ Enable
☒ Disable

Staged cooling

☐ Enable
☒ Disable

DeltaT lockouts

☐ Enable
☒ Disable

Humidification

☐ Enable
☒ Disable

Dehumidification

☐ Enable
☒ Disable

Staged heat mixed air humidity

No limit

Staged heat outdoor air temperature

No limit

Staged heat outdoor air humidity

No limit

Staged cool mixed air temperature

No limit

Staged cool mixed air humidity

No limit

Staged cool outdoor air temperature

No limit

Staged cool outdoor air humidity

No limit

SAVE

BUILD POINTS

The space temperature setpoints entered must be as follows:
unoccupied heat setpoint ≤ standby heat setpoint ≤ occupied heat setpoint < occupied cool setpoint ≤ standby cool setpoint ≤ unoccupied cool setpoint.

If a setpoint was entered out of range, when attempting to save the configuration, an error message displays.

SETPOINTS - TC300

This Menu allows the user to configure the setpoints for different modes like occupied, standby, and unoccupied.

Honeywell | TC300 Thermostat Configuration Wizard

 Configuration wizard ▾

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Setpoint Option

Occupied Heat Setpoint

20

°C[4.5, 32.0]

Occupied Cool Setpoint

24.5

°C[10.0, 37.0]

Standby Heat Setpoint

18.5

°C[4.5, 32.0]

Standby Cool Setpoint

26.5

°C[10.0, 37.0]

OAT at max. Setpoint Temp

21

°C[-40, 49]

OAT at min. Setpoint Temp

32

°C[-40, 49]

Heating Recovery

Max. Setpoint Ramp

4.5

°C/hr[0, 20]

Min. Setpoint Ramp

1

°C/hr[0, 20]

OAT at max. Setpoint Temp

15.5

°C[-40, 49]

OAT at min. Setpoint Temp

-18

°C[-40, 49]

SAVE

Fig. 10 Setpoints

The space temperature setpoints entered must be as follows:
unoccupied heat setpoint ≤ Occupied Cool setpoint ≤ Standby heat setpoint < Standby cool setpoint ≤ Unoccupied Heat setpoint ≤ unoccupied cool setpoint.

 **NOTE:**
If a setpoint was entered out of range, when attempting to save the configuration, an error message displays.

PID CONFIGURATIONS - TC500A

This Menu allows the user to configure the PID settings for cooling and heating options.

Honeywell

| TC500A Thermostat Configuration Wizard

 Configuration wizard ▾

General

Equipment

Control Loops

Setpoints

PID Config.

Fan Modes

Sensors

Terminal Assignment

Cooling options

OAT cooling lockout setpoint

35.0

°F[-40.0-120.0]

DAT cooling low limit

45.0

°F[-40.0-60.0]

Throttling range mode

☒ Auto

☐ Manual

Heating options

OAT heating lockout setpoint

65.0

°F[-40.0-120.0]

DAT heating high limit

140.0

°F[65.0-140.0]

Throttling range mode

☒ Auto

☐ Manual

SAVE

BUILD POINTS

Fig. 11 PID Configurations

PID CONFIGURATIONS - TC300

This Menu allows the user to configure the PID settings for cooling and heating options.

TC300 Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Cooling Options

DAT Low Limit

45

°F[-40, 60]

Cooling Throttling Range

4

Δ°F[0, 30]

Cooling Integral Time

2500

s[0, 5000]

Heating Options

DAT High Limit

150

°F[60, 200]

Heating Throttling Range

4

Δ°F[0, 30]

Heating Integral Time

2500

s[0, 5000]

SAVE

Fig. 12 PID Configurations

FAN MODES - TC500A

The fan modes Menu allows the user to set parameters for the fan. The below settings can be changed.

Honeywell

TC500A Thermostat Configuration Wizard

Configuration wizard ▾

General

Equipment

Control Loops

Setpoints

PID Config.

Fan Modes

Sensors

Terminal Assignment

Fan options

Fan mode ⓘ

Continuous ▾

Fan extended run time in heat

90.0

s[0.0-300.0]

Fan extended run time in cool

0.0

s[0.0-300.0]

Enable fan run with heat

☒ Enable

☐ Disable

Speed type ⓘ

Single speed ▾

SAVE

BUILD POINTS

Fig. 13 Fan Modes

FAN TYPE - TC300

The fan modes Menu allows the user to set parameters for the fan. The below settings can be changed.

Honeywell

TC300 Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Fan

Fan Type

Variable Speed

Variable Speed Type

0-10v

2-10v

Cooling Mode Min Speed

20

%[0, 100]

Cooling Mode Max Speed

100

%[0, 100]

Heating Mode Min Speed

10

%[0, 100]

Heating Mode Max Speed

50

%[0, 100]

Ventilation Mode Fan Speed

20

%[0, 100]

Fan Delay

Cooling Fan Off Delay Time

0

s[0, 180]

Heating Fan On Delay Time

30

s[0, 30]

Heating Fan Off Delay Time

120

s[0, 180]

Default Fan Mode

Default Fan Mode

Auto

SAVE

Fig. 14 Fan Modes

NOTE:

Applicable only for TC300 Device.

SENSORS - TC500A

This Menu allows the user to configure the sensors.

Honeywell

TC500A Thermostat Configuration Wizard

Configuration wizard ▾

General

Equipment

Control Loops

Setpoints

PID Config.

Fan Modes

Sensors

Terminal Assignment

Control sensor

Main control sensor

Local temp humidity ▾

Internal temperature offset

0.0

°F[-10.0-10.0]

Internal humidity offset

0.0

%[-10.0-10.0]

Input characteristics

Occupancy sensor check

☒ Direct

☐ Reverse

Proof of water flow

☒ Direct

☐ Reverse

Dirty filter check

☒ Direct

☐ Reverse

Airflow status check

☒ Direct

☐ Reverse

Shutdown check

☒ Direct

☐ Reverse

CO2 sensor type

☐ 0-10V

☒ 2-10V

Window open sensor check

☒ Direct

☐ Reverse

Drain pan sensor

☒ Normally open

☐ Normally closed

Mixed air sensor

NTC 20 K ▾

SAVE

BUILD POINTS

Fig. 15 Sensors

SENSORS - TC300

This Menu allows the user to configure the sensors.

Honeywell

TC300 Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Sensors

Occupancy

Occupancy Sensor Type

Normally Open

Normally Closed

Proof Of Airflow

Proof Of Airflow Sensor Type

Normally Open

Normally Closed

Discharge Air Temperature

Discharge Air Sensor Type

NTC20K

Discharge Temperature High Limit

140

°F[70, 180]

Discharge Temperature Low Limit

45

°F[35, 65]

Space Temperature

Space Temperature Sensor Type

NTC20K

Space Temperature High Limit

90

°F[90, 150]

Space Temperature Low Limit

45

°F[0, 60]

Drain Pan

Drain Pan Sensor Type

Normally Open

Normally Closed

Outdoor Air Temperature

Shutdown

Custom Sensor

Custom Sensor 1

Custom Sensor Type

Digital Input

Digital Input

Normally Open

Normally Closed

Custom Sensor 2

Custom Sensor 3

SAVE

Fig. 16 Sensors

TERMINAL ASSIGNMENT - TC500A

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings.

Honeywell

TC500A Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Control Loops

Setpoints

PID Config.

Fan Modes

Sensors

Terminal Assignment

G	D01	Fan command
W1	D02	Heating stage1 command
W2	D03	Heating stage2 command
W3	D04	None
Y1	D05	Cooling / Compressor stage1 comm
Y2	D06	Cooling / Compressor stage2 comm
Y3	D07	None
	D08	None
UI01	UI01	None
UI02	UI02	None
UI1	UI1	None
UI2	UI2	None

SAVE

BUILD POINTS

Fig. 17 Terminal Assignment

Table 4 Terminal Assignment

Terminal	Options
D01	1: None 2: Fan command (default) 3: Fan high-speed command
D02	1: Heating stage1 command (default)
D03	1: Heating stage2 command (default)
D04	1: None (default) 2: Heating stage3 command 3: Heat pump reversing valve command 4: Fan low-speed command 5: Occupancy status 6: Dehumidification command 7: Humidification command 8: Purge output
D05	1: Cooling / Compressor stage1 command (default)
D06	1: Cooling / Compressor stage2 command (default)
D07	1: None (default) 2: Cooling / Compressor stage3 command 3: Economizer min. damper command 4: Fan low-speed command 5: Occupancy status 6: Dehumidification command 7: Humidification command
D08	1: None (default) 2: Economizer min. damper command 3: Fan low-speed command 4: Occupancy status 5: Dehumidification command 6: Humidification command 7: Purge output
UIO1	1: None (default) 2: Occupancy sensor 3: Dirty filter 4: Air flow status 5: Shutdown 6: Mixed air sensor 7: Outside air sensor 8: Discharge air sensor 9: CO2 sensor 10: Fan speed control 11: Water flow status 12: Space temperature sensor 13: Packaged economizer fault 14: Window open 15: Return air sensor 16: CO2 output 17: Purge output

Table 4 Terminal Assignment (Continued)

Terminal	Options
UIO2	1: None (default) 2: Occupancy sensor 3: Dirty filter 4: Air flow status 5: Shutdown 6: Mixed air sensor 7: Outside air sensor 8: Discharge air sensor 9: CO2 sensor 10: Heating control 11: Water flow status 12: Space temperature sensor 13: Packaged economizer fault 14: Window open 15: Return air sensor 16: CO2 output 17: Purge output
UI1	1: None (default) 2: Occupancy sensor 3: Dirty filter 4: Air flow status 5: Shutdown 6: Mixed air sensor 7: Outside air sensor 8: Discharge air sensor 9: CO2 sensor 10: Water flow status 11: Space temperature sensor 12: Packaged economizer fault 13: Window open 14: Return air sensor
UI2	1: None (default) 2: Occupancy sensor 3: Dirty filter 4: Air flow status 5: Shutdown 6: Mixed air sensor 7: Outside air sensor 8: Discharge air sensor 9: CO2 sensor 10: Water flow status 11: Space temperature sensor 12: Heat pump frequency 13: Packaged economizer fault 14: Window open 15: Return air sensor

TERMINAL ASSIGNMENT - TC300

This Menu allows the user to configure the different UIs, UIOs, or DOs based on the equipment and the other settings.

Honeywell

TC300 Thermostat Configuration Wizard

Configuration wizard

General

Equipment

Setpoint

Fan Type

PID Config

Sensors

Terminal Assignment

Equipment

Heating On/Off

DO1

Cooling On/Off

DO2

Fan

Variable Speed Fan

UIO1

Sensors

Occupancy Sensor

Select One Item

Proof Of Airflow Sensor

Select One Item

Discharge Air Sensor

Select One Item

Space Temp Sensor

Select One Item

Drain Pan Sensor

Select One Item

SAVE

Fig. 18 Terminal Assignment

Table 5 Terminal Assignment

Terminal	Options
DO1	1: NA 2: On/Off Heat (default) 3: Heating On/Off 4: Heating Floating Open 5: Cooling Floating Open 6: Valve On/Off 7: Valve Floating Open 8: Changeover Valve 9: fan Command 10: High Speed Fan 11: Medium Speed Fan 12: Low Speed Fan 13: Auxiliary Heat 14: Heat Stage 1 15: Valve Stage 1 Note: Changeover valve used to switch between heating and cooling mode.
DO2	1: NA 2: On/Off Cool (default) 3: Heating Floating Close 4: Cool Floating Close 5: Cooling On/Off 6: Valve Floating Close 7: Changeover Valve 8: Fan Command 9: High Speed Fan 10: Medium Speed Fan 11: Low Speed Fan 12: Auxiliary Heat 13: Cool Stage 1
DO3	1: NA 2: NA (default) 3: Cooling Floating Open 4: Changeover Valve 5: Fan Command 6: High Speed Fan 7: Medium Speed Fan 8: Low Speed Fan 9: Auxiliary Heat 10: Heat Stage 1 11: Cool Stage 1

Table 5 Terminal Assignment (Continued)

Terminal	Options
DIO1	1: NA (default) Input 1: Discharge Air Sensor 2: Drain Pan Sensor 3: Occupancy Sensor 4: Proof of Airflow 5: Pipe Sensor 6: Space Temp Sensor 7: Changeover Switch Output 1: Cooling Floating Close 2: Changeover Valve 3: Fan Command 4: High Speed Fan 5: Medium Speed Fan 6: Low Speed Fan 7: Auxiliary Heat
DIO2	1: NA (default) Input 1: Discharge Air Sensor 2: Drain Pan Sensor 3: Occupancy Sensor 4: Proof of Airflow 5: Pipe Sensor 6: Space Temp Sensor 7: Changeover Switch Output 1: Changeover Valve 2: Fan Command 3: High Speed Fan 4: Medium Speed Fan 5: Low Speed Fan 6: Auxiliary Heat
UIO1	1: NA (default) Input 1: Discharge Air Sensor 2: Drain Pan Sensor 3: Occupancy Sensor 4: Proof of Airflow 5: Pipe Sensor 6: Space Temp Sensor 7: Changeover Switch Output 1: 6-Way Valve 2: Modulating Cool 3: Modulating Heat 4: Modulating Valve 5: UIO1 UIO2 NA Variable Speed Fan

Table 5 Terminal Assignment (Continued)

Terminal	Options
UIO2	1: NA (default) Input 1: Discharge Air Sensor 2: Drain Pan Sensor 3: Occupancy Sensor 4: Proof of Airflow 5: Pipe Sensor 6: Space Temp Sensor 7: Changeover Switch Output 1: 6-Way Valve 2: Modulating Cool 3: Modulating Heat 4: Modulating Valve 5: UIO1 UIO2 NA Variable Speed Fan
UIO3	1: NA (default) Input 1: Discharge Air Sensor 2: Drain Pan Sensor 3: Occupancy Sensor 4: Proof of Airflow 5: Pipe Sensor 6: Space Temp Sensor 7: Changeover Switch Output 1: 6-Way Valve 2: Modulating Cool 3: Modulating Heat 4: Modulating Valve 5: UIO1 UIO2 NA Variable Speed Fan

UPLOAD/DOWNLOAD CONFIGURATION

The TC Series Thermostat Configuration Wizard allows the user to download and upload the configuration to the TC Series device after configuration changes.

Download Configuration

This action downloads the saved configuration from the wizard to the TC500A device. After every configuration change, it is essential to download the configuration to the device to reflect the new settings. To download the updated configuration to the TC Series thermostat device, right-click on the TC500 device, click **Actions**, and click **Download Configuration**

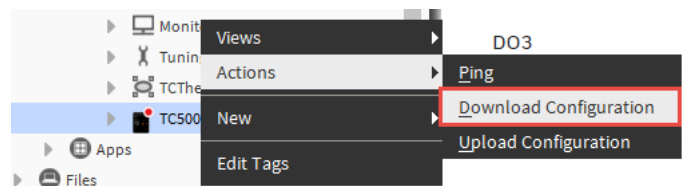


Fig. 19 Download Configuration

Download Configuration (UX Screen)

Double click on Configuration Download

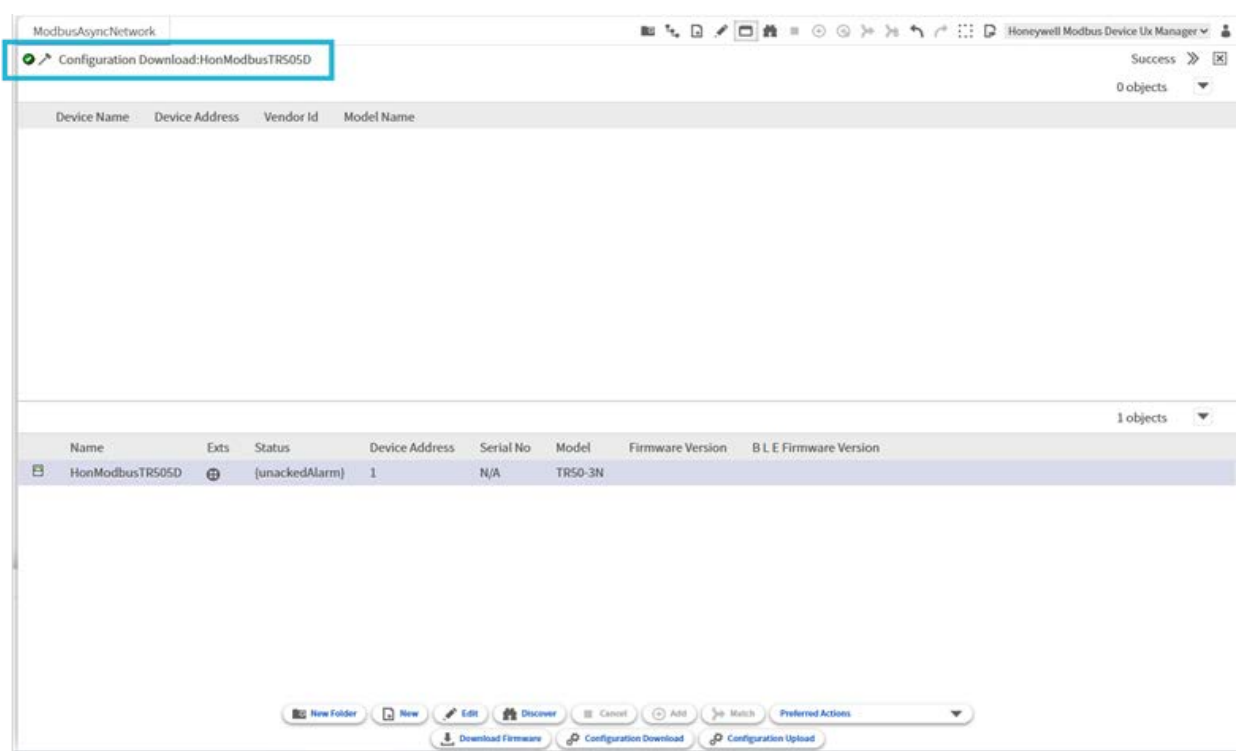


Fig. 20 Download Configuration (UX screen)

Upload Configuration

This action allows the user to upload the configuration saved in the TC Series device. After uploading, settings can be reconfigured and downloaded to the device again. To upload the existing configuration from the TC Series device, click **Actions** and then click **Upload Configuration**.



Fig. 21 Upload Configuration

Upload Configuration (UX Screen)

Double click on Configuration Download .

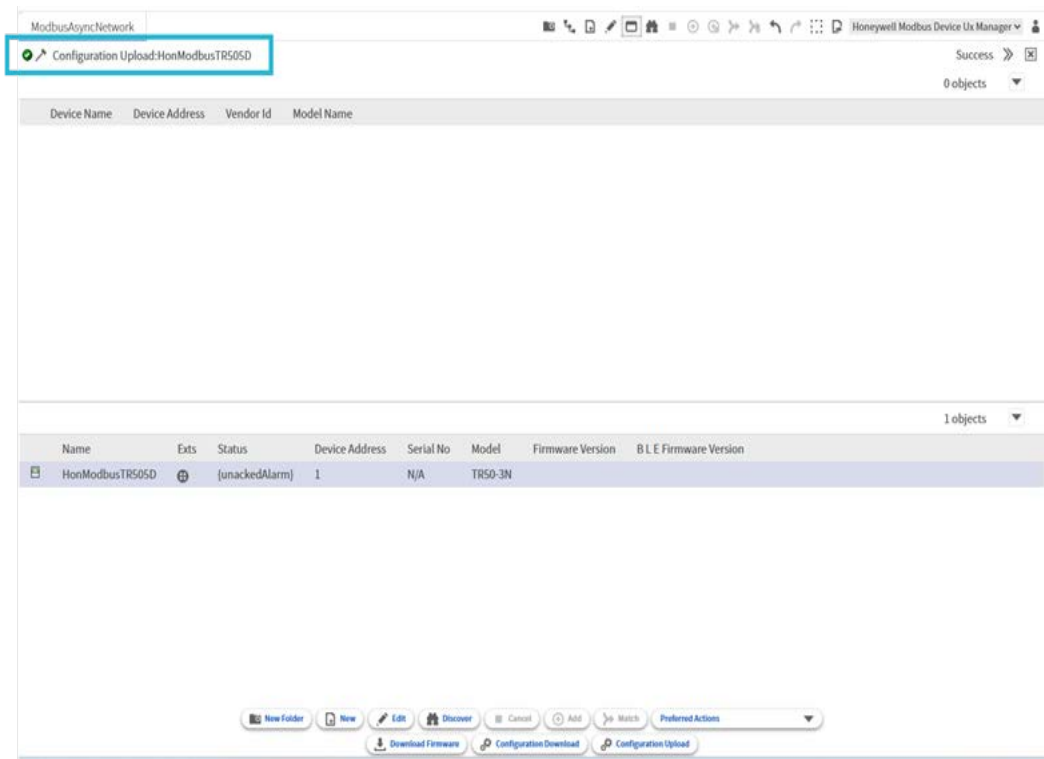


Fig. 22 Upload Configuration (UX Screen)



NOTE:

The upload and download configuration will support undefined values only when the thermostat has valid enum ordinal.

SERVICE MODE - TC500A

Service mode allows the user to test the output of the equipment for the mechanical operation.

Click on the drop-down menu and select the service mode to navigate to the service mode page. The service mode includes the analog and digital outputs of the TC500A thermostat. The output points display the outputs available on the thermostat, and the override option allows to configure a value and set the output on the thermostat.

To use the service mode:

1. Configure the toggle switch to ON to enable the override value fields.

Honeywell

| TC500A Thermostat Configuration Wizard

Configuration wizard ▾
Configuration wizard
Service mode
Alarm view

Service mode

Output points

Fan

Fan speed

Heat stage1

Heat stage2

Heat stage3

Modulating heating

Heat pump reversing valve

Cooling stage1

Cooling stage2

Cooling stage3

Humidification

Dehumidification

Economizer min. damper

Outside air damper

Occupancy command

Purge output 2V-10V

CO2 output

Override

Off ▾ On/Off

0 [0-100%]

Off ▾ On/Off

Off ▾ On/Off

Off ▾ On/Off

0 [0-100%]

Off ▾ On/Off

Off ▾ On/Off

Off ▾ On/Off

Off ▾ On/Off

Off ▾ On/Off

Off ▾ On/Off

0 [0-100%]

Off ▾ On/Off

0 [0-100%]

0 [0-100%]

SET

Fig. 23 Service Mode - Disabled

2. Click **Yes** on the notification displayed to enable the service mode.

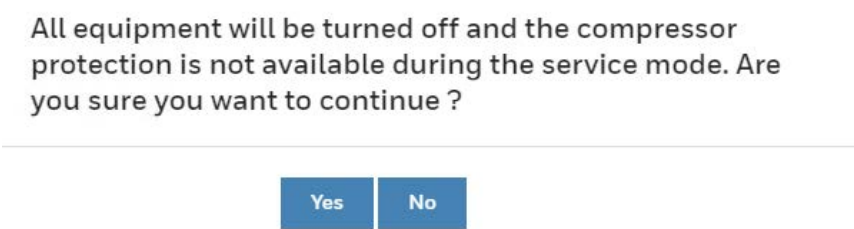


Fig. 24 Dialogue Box to Enable the Service Mode

3. Configure the required values in the override fields and click on the set button to set the values to the output on the thermostat.

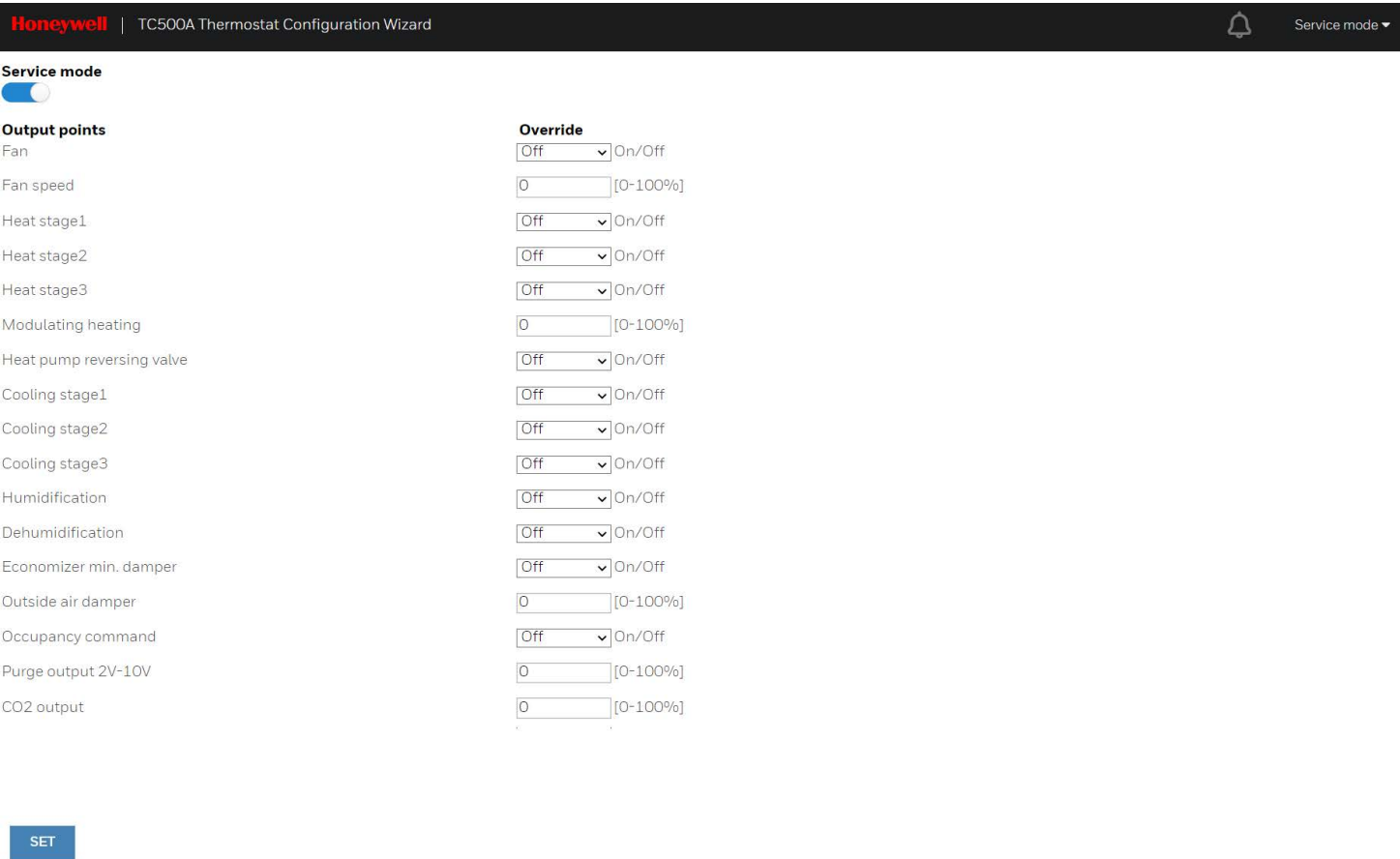


Fig. 25 Service Mode - Enabled

1. Disable the toggle switch to disable the service mode.
2. Click on the drop-down menu and select the configuration wizard to navigate to the configuration wizard view. A prompt to exit the service mode will appear. Select Yes, if you want to exit the service mode and navigate to the configuration wizard. Select No, if you want to continue with the overridden points and navigate to the configuration wizard. Select cancel to continue with the service mode and not navigate to the configuration wizard.

There are outputs set to manual mode. Do you want to put the outputs to the auto mode?

Yes

No

Cancel

Fig. 26 Dialogue Box to Disable the Service Mode



NOTE:

The service mode is displayed only when the device is online. The outputs are written on the network inputs. Use the respective network output to validate the point value.

The override is for a one-hour duration and restarts whenever a point is overridden using set option. After one-hour, the thermostat will exit service mode.

SERVICE MODE - TC300

Service mode allows the user to test the output of the equipment for the mechanical operation.

Click on the drop-down menu and select the service mode to navigate to the service mode page. The service mode includes the analog and digital outputs of the TC300 thermostat. The output points display the outputs available on the thermostat, and the override option allows to configure a value and set the output on the thermostat.

To use the service mode:

1. Configure the toggle switch to ON to enable the override value fields.

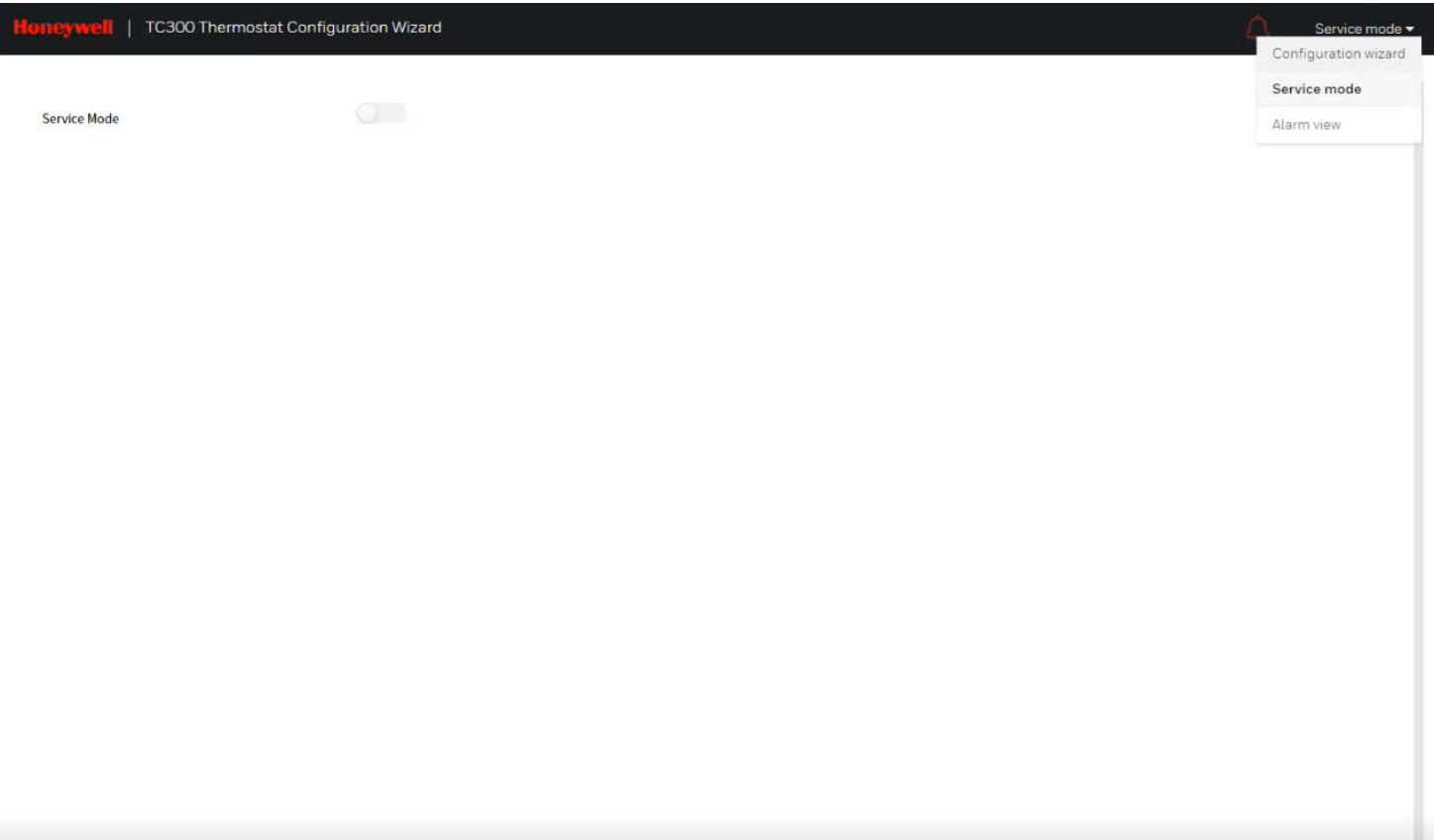


Fig. 27 Service Mode - Disabled

2. Click **Yes** on the notification displayed to enable the service mode.

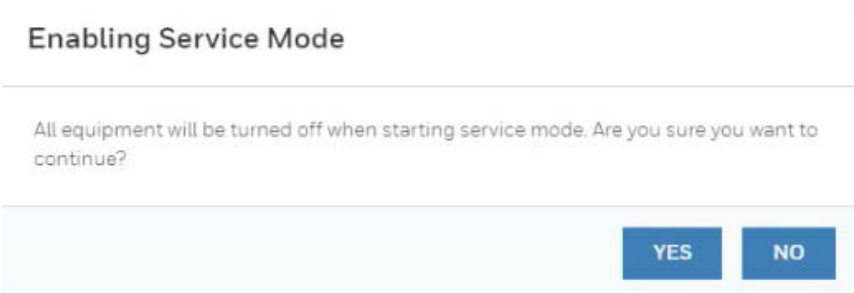


Fig. 28 Dialogue Box to Enable the Service Mode

3. Configure the required values in the override fields and click on the set button to set the values to the output on the thermostat.

The screenshot shows the Honeywell TC300 Thermostat Configuration Wizard interface. At the top, there is a header bar with the Honeywell logo and the text "TC300 Thermostat Configuration Wizard". On the right side of the header, there is a bell icon and a dropdown menu labeled "Service mode". Below the header, the "Service Mode" toggle switch is turned on (blue). Under the "Fan" section, the "Variable Speed Fan" is set to 0, with a range of 0 to 100. Under the "Equipment" section, the "Heating On/Off" and "Cooling On/Off" toggle switches are both turned off (grey).

Fig. 29 Service Mode - Enabled

1. Disable the toggle switch to disable the service mode.
2. Click on the drop-down menu and select the configuration wizard to navigate to the configuration wizard view. A prompt to exit the service mode will appear. Select Yes, if you want to exit the service mode and navigate to the configuration wizard. Select No, if you want to continue with the overridden points and navigate to the configuration wizard. Select cancel to continue with the service mode and not navigate to the configuration wizard.

The screenshot shows a dialog box titled "Exiting Service Mode". The text inside the dialog box reads: "Exiting service mode will resume normal thermostat function. Do you want to exit service mode?". At the bottom right of the dialog box, there are two buttons: "YES" and "NO".

Fig. 30 Dialogue Box to Disable the Service Mode

 NOTE:

The service mode is displayed only when the device is online. The outputs are written on the network inputs. Use the respective network output to validate the point value.

The override is for a one-hour duration and restarts whenever a point is overridden using set option. After one-hour, the thermostat will exit service mode.

ALARM VIEW - TC500A

The alarm view allows the user to monitor the predefined alarms in the TC500A thermostat. Navigate to the alarm view page from the navigation tree of the wizard. A red color alarm icon on the header indicates an active alarm in the thermostat, and a white color alarm icon indicates no alarm. The alarm priority is displayed for a point in the alarm. Hover over the tooltip to view the alarm description.

Honeywell TC500A Thermostat Configuration Wizard		 Alarm view ▼
Alarms		
Proof of air flow alarm	Normal	
Space freeze protection alarm	Normal	
Proof of water flow alarm	Normal	
Space temperature sensor failure	Medium priority alarm	
Space humidity sensor failure	Medium priority alarm	
Space temperature out of range alarm	Normal	
Space CO2 limit exceeded	Normal	
Shutdown alarm	Normal	
Mixed air sensor failure alarm	Normal	
Outdoor air temperature sensor failure	Normal	
Discharge air temperature sensor failure	Normal	
Space CO2 sensor fault	Normal	
Water flow fault	Normal	
Sylk device communication failure	Many sytk fail alarm	
Unknown time	Normal	
Internal proximity sensor failure	Normal	
External economizer fault	Normal	
Window open detection alarm	Normal	
Delta T out of range alarm	Normal	
Return air sensor failure	Normal	
Coil freeze protection alarm	Normal	
Internal economizer fault	Normal	

Fig. 31 Alarm View

The following points are configured in the alarm view.

Table 6 Alarm View

Alarm Parameters	Alarm Priority
Proof of air flow alarm	High priority alarm
Space freeze protection alarm	High priority alarm
Proof of water flow alarm	High priority alarm
Space temperature sensor failure	Medium priority alarm High priority alarm
Space humidity sensor failure	Medium priority alarm High priority alarm
Space temperature out of range	Medium priority alarm
Space CO2 limit exceeded	Medium priority alarm (CO2 threshold limit exceeded)
Shutdown alarm	Medium priority alarm
Mixed air sensor failure alarm	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Outdoor air temperature sensor failure	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Discharge air temperature sensor failure alarm	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Space CO2 sensor fault	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Water flow fault	Medium priority alarm
Sylk device communication failure	Sylk address 2 fail alarm Sylk address 3 fail alarm Sylk address 4 fail alarm Sylk address 5 fail alarm Sylk address 6 fail alarm Unconfigured Sylk address 8 fail alarm Sylk address 9 fail alarm Sylk address 10 fail alarm Sylk address 11 fail alarm Sylk address 12 fail alarm Many Sylk fail alarm
Unknown time	High priority alarm
Internal proximity sensor failure	High priority alarm
External economizer fault	High priority alarm
Window open detection alarm	Medium priority alarm
Delta T out of range alarm	Medium priority alarm High priority alarm
Return Air sensor failure	Medium priority alarm (Out of range) High priority alarm (Sensor failure)
Coil freeze protection alarm	High priority alarm

Table 6 Alarm View (Continued)

Alarm Parameters	Alarm Priority
Internal economizer fault	Damper stuck open alarm Damper stuck at minimum alarm Bad or unplugged actuator alarm Actuator mechanically disconnected alarm
Drain pan sensor alarm	Normal alarm High Priority alarm

ALARM VIEW - TC300

The alarm view allows the user to monitor the predefined alarms in the TC300 thermostat. Navigate to the alarm view page from the navigation tree of the wizard. A red color alarm icon on the header indicates an active alarm in the thermostat, and a white color alarm icon indicates no alarm. The alarm priority is displayed for a point in the alarm. Hover over the tool tip to view the alarm description.

Honeywell TC300 Thermostat Configuration Wizard		Alarm view ▾
Alarms		
Proof of Air Flow Alarm	Inactive	
Space Freeze Protection Alarm	Inactive	
Discharge Air Temperature Sensor Failure	Inactive	
Space Temperature Sensor Failure ⓘ	Active(High)	
Sylk Device Communication Failure ⓘ	Active(High)	
Space Humidity Sensor Failure ⓘ	Active(High)	
Unknown Time ⓘ	Active(Medium)	
Space Temperature Out Of Range	Inactive	
Drain Pan Sensor Alarm	Inactive	
Pipe Sensor Failure	Inactive	
Water Temperature Is Not Suitable For Heating/Cooling	Inactive	
Room Temperature Changing Trend is Reversed With System Mode	Inactive	
Pipe Sensor Out of Range	Inactive	

Fig. 32 Alarm View

List of Alarms and their Severity

The list of alarms in the Commercial Connected thermostat is as follows

Table 7 List of Alarm and their Severity

Alarms	Severity
Proof of Air Flow Alarm	High
Space Freeze Protection Alarm	High
Space Temperature Sensor Failure	High/Medium
Space Temperature Out of Range	High/Medium
Space Humidity Sensor Failure	High/Medium
Discharge Air Temp. Sensor Failure	High
Discharge Air Temperature out of range alarm	Medium
Drain Pan Sensor Alarm	High
Sylk Device Communication Failure	High
Pipe Sensor Failure	High
Pipe Sensor Out of Range	High

Table 7 List of Alarm and their Severity (Continued)

Alarms	Severity
Water Temperature is Not Suitable for Heating/Cooling	High
Room Temperature Changing Trend is Reversed with System Mode	High
Unknown Time	Medium

FIRMWARE DOWNLOAD TC SERIES

NOTE:

TC500A and TC300 Configuration are same but the TC500A configuration is shown as reference.

1. Expand **Station > Config > Driver**.
2. Right-click on the BacnetNetwork > Views > T C Thermostat Bacnet Device Manager.
The list of all available Bacnet devices which are in the database appears.

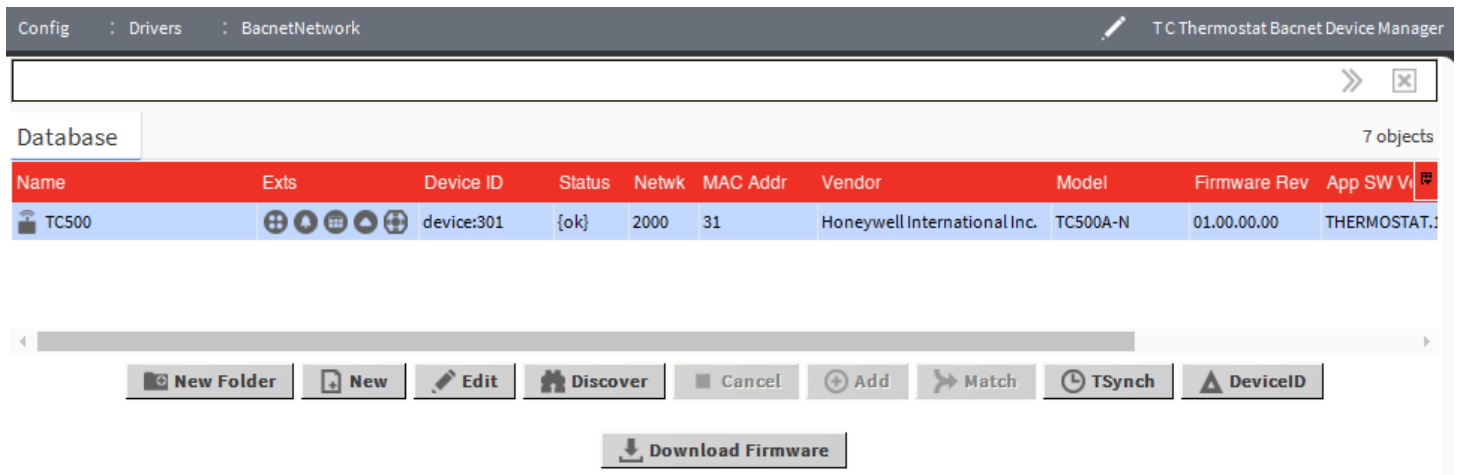


Fig. 33 Firmware Download

3. Select the TC500A thermostat device (Device type = honeywellTCThermostatWizard:TC500).
4. Click on the **Download Firmware** button. The file chooser window appears.
5. Select the firmware file and click on the **Open** button.
The firmware download starts, and the user will get a notification after the download process is completed.

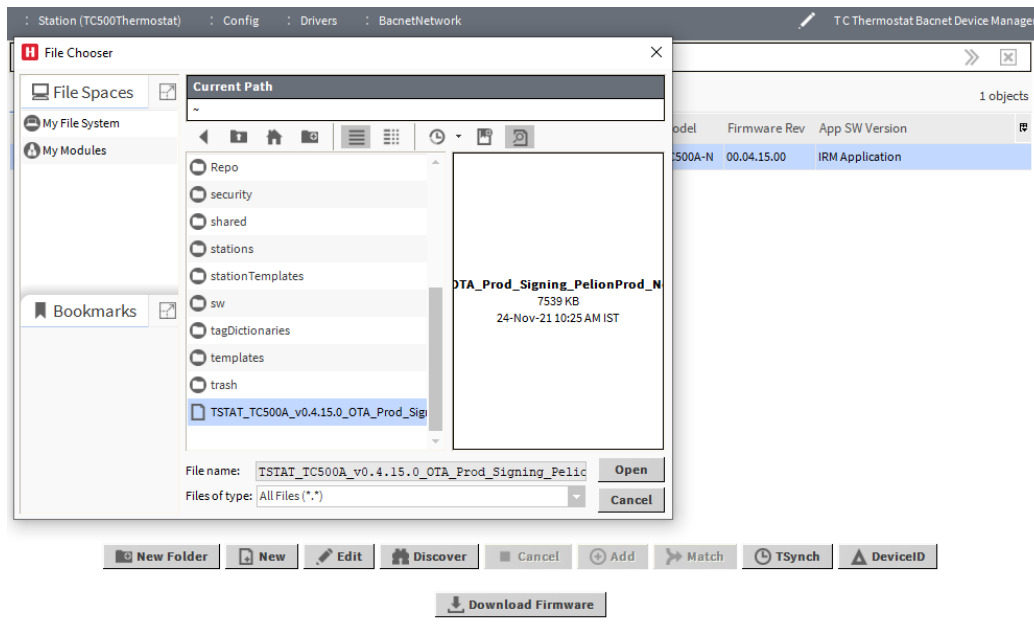


Fig. 34 File Chooser

6. Select the firmware file and click on ok button.

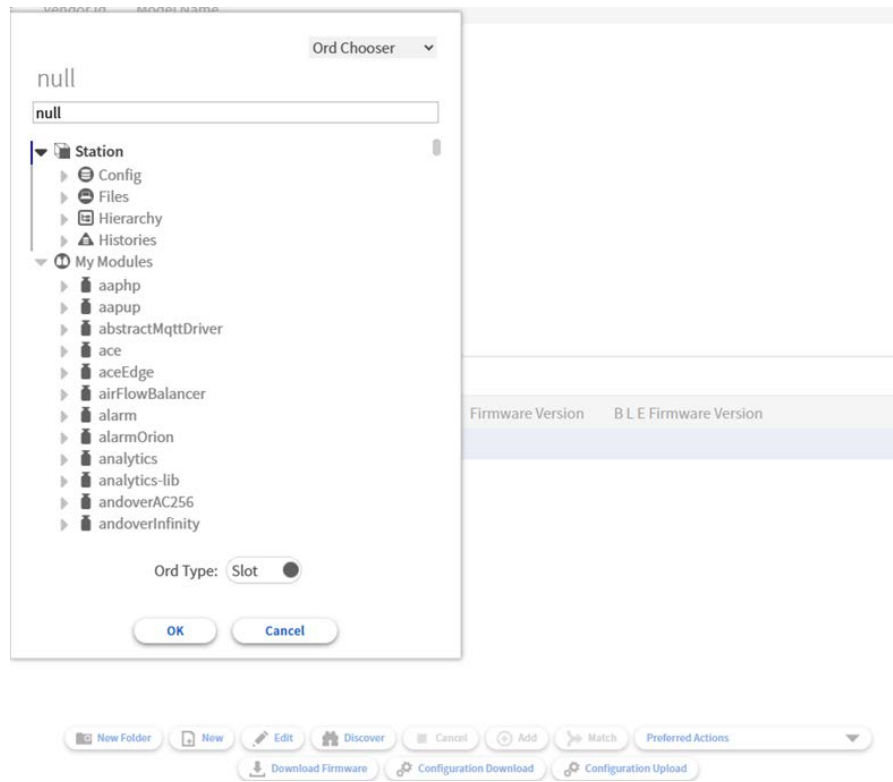


Fig. 35 File Chooser (UX Screen)

**NOTE:**

The firmware download is supported only through the workbench. Refresh the device manager view page when the firmware download is finished to verify if the firmware version number has been updated.

APPLICABLE TECHNICAL LITERATURE

Table 8 Applicable Technical Literature

Document Name	Document Number
TC500A Thermostat BACnet Integration Guide	31-00478
TC Thermostat Configuration Wizard Installation Instructions	31-00489
TC500A Commercial Thermostat Configuration and User Guide	31-00400M
TC300 Thermostat BACnet Integration Guide	31-00646

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31-00490-03I Rev. 03-26

